

Home Assignment:

Test-Case Quality Gate and Publishing Workflow

Enhance the existing test-case generator application by adding a quality-validation and approval workflow.

Evaluate every generated test case using these three DeepEval metrics:

- Faithfulness: Score must be ≥ 0.75
- Contextual Recall: Score must be ≥ 0.75
- Hallucination: Score must be ≤ 0.25 , preferably 0

Add a toggle to support two modes:

1. Manual Mode

- Display all three metric scores in the UI.
- Provide Approve and Reject buttons.
- When the user approves the test cases, store them in MongoDB and create them in Jira.
- When the user rejects them, do not publish or store the test cases.

2. Automation Mode

- Automatically validate the generated test cases against the configured thresholds.
- Publish them to MongoDB and Jira only when all three metrics pass.
- If any metric fails, stop the publishing process and display the failure reason.

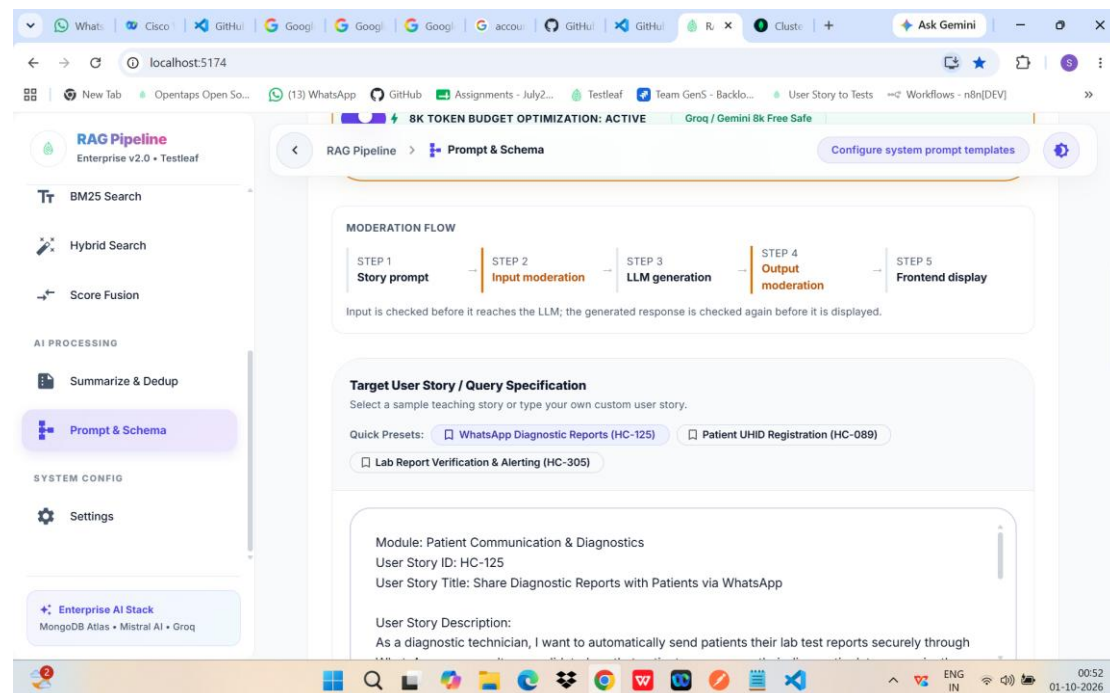
Nice-to-Have Enhancement

Extend the publishing integration to support additional test-management tools:

- AIO Tests
- Xray
- Zephyr

Test the complete workflow in both Manual and Automation modes and submit the metric results, and final execution screenshots.

Note : The auto-review compares mismatched reference and context data, which can legitimately produce zero. It uses each newly generated test case's expected result as expected_output; in baseline mode, it uses the user story itself as context. In RAG mode, it uses retrieved cases and a generated summary as context. Contextual Recall asks whether the context supports the expected answer, so a generated expected result that isn't stated in those sources can score zero - so made **Contextual Recall is informational**



Browser tabs: (118) Vi, RAG Pl, System, GenAIS, New To, Cluster, Context, Context, Ask Gemini

Address bar: samemmanuel1987.atlassian.net/jira/software/projects/GN/boards/34/backlog?selectedIssue=GN-32

Navigation: New Tab, Opentaps Open So..., (13) WhatsApp, GitHub, Assignments - July2..., Testleaf, Team GenS - Backdo..., User Story to Tests, Workflows - n8n[DEV]

Jira Search: Search

Buttons: + Create, Upgrade

Left sidebar: For you, Recent, Starred, Apps, Plans, Spaces, Recent, GenAIS, More spaces, Recommended, Collect requests, Filters, Dashboards, Assets, Teams, Goals, Projects

Spaces: GenAIS

Summary, List, Board, Backlog, More 5 +

Search backlog, Filter

Issue	Summary	Status
GN-26	The application failed to redir...	To Do
GN-27	The application failed to redir...	To Do
GN-28	The application failed to redir...	To Do
GN-29	Auto-Refresh Queue List Withi...	To Do
GN-30	Emergency Filter Shows Only ...	To Do
GN-31	Scroll Position and Active Filte...	To Do
GN-32	Negative & Edge Cases: Empt...	To Do

+ Create

Jira work item

Description: Validate system behavior when a filter yields no patients and when a patient's status changes multiple times within a single refresh interval.

Preconditions: Doctor dashboard open; no patients currently have status 'Discharged' in the ward; ability to modify patient status via admin console.

Steps:

1. Login as a doctor and navigate to the AHC Queue List.
2. Apply the filter 'Status = Discharged'.
3. Verify that the queue displays a friendly empty-state message (e.g., 'No discharged patients').
4. Using the admin console, change Patient A's status from 'Waiting' → 'Under Examination' → 'Discharged' within 10 seconds.

Browser tabs: What, Cisco, GitHub, Google, Google, Google, accou, GitHub, RAG F, CI, Ask Gemini

Address bar: cloud.mongodb.com/v2/6a802f260e1ad45327638dbf#explorer/6a802f77669c9eb12f3ed1f/RAG_testCases/testcases/find

Navigation: New Tab, Opentaps Open So..., (13) WhatsApp, GitHub, Assignments - July2..., Testleaf, Team GenS - Backdo..., User Story to Tests, Workflows - n8n[DEV]

Organization: Sam's Org - 2026-08-15

Project: Project 0

Buttons: Compare tiers

Data Explorer

My Queries, Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

testcases

Documents 6.4K, Aggregations, Schema, Indexes 1, Validation

Type a query: { field: 'value' } or Generate query

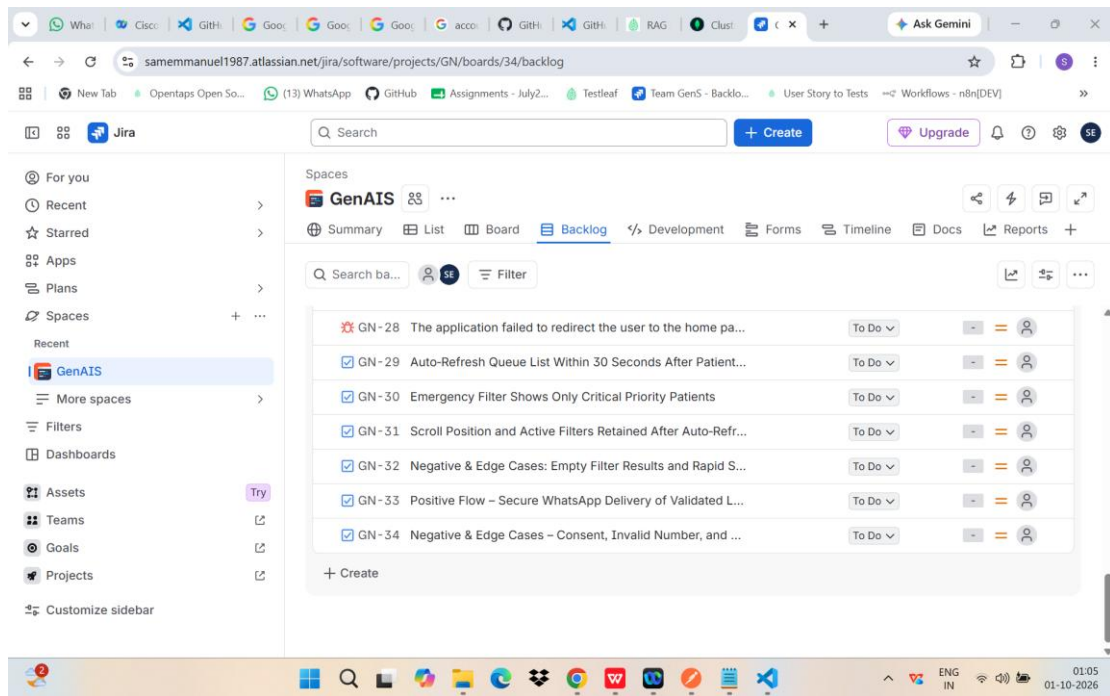
Buttons: Explain, Reset, Find, Options

100, 1 - 100 of 6358

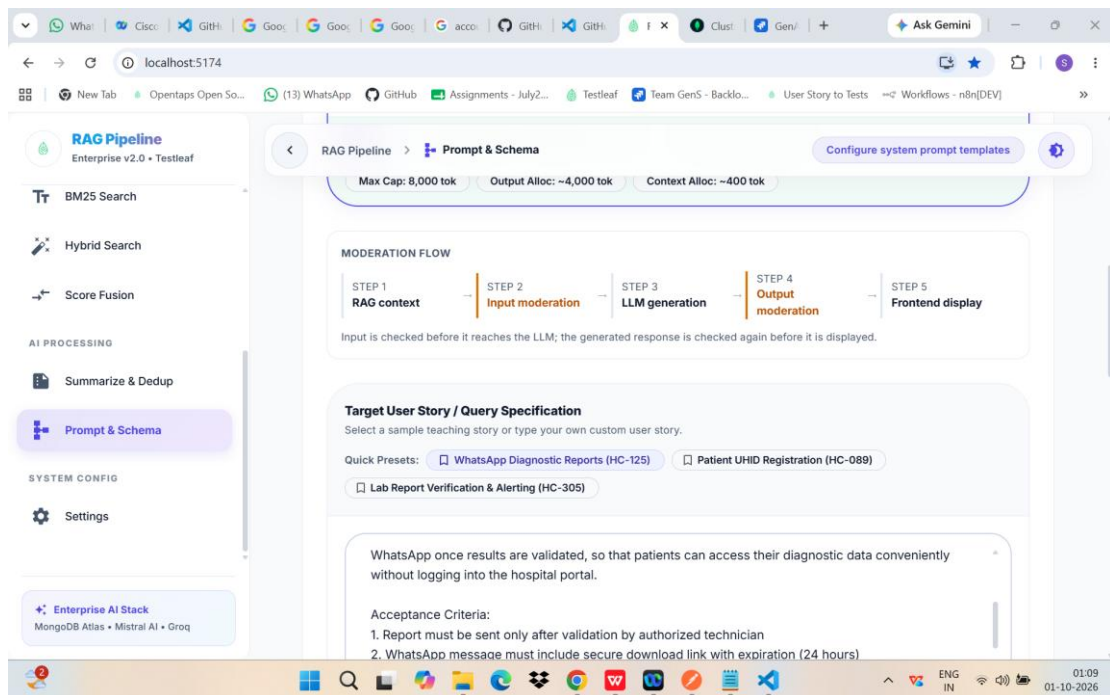
```
{
  "_id": ObjectId("6a820126e13415c9ef066fcb"),
  "module": "Registration",
  "id": "TC_005",
  "preRequisites": "PRN/ERN Number",
  "title": "Conversion from PRN/ERN Reg to UHID",
  "description": "Verify Conversion from PRN/ERN Reg to UHID",
  "steps": "1.Launch the application\n2.Login to the application with valid credentials.\n3.Click on Patient Service Tab and select Registration from the drop down...",
  "expectedResults": "Should get convert successfully from PRN/ERN Reg to UHID",
  "automationManual": "Automated",
  "priority": "P1"
}
```

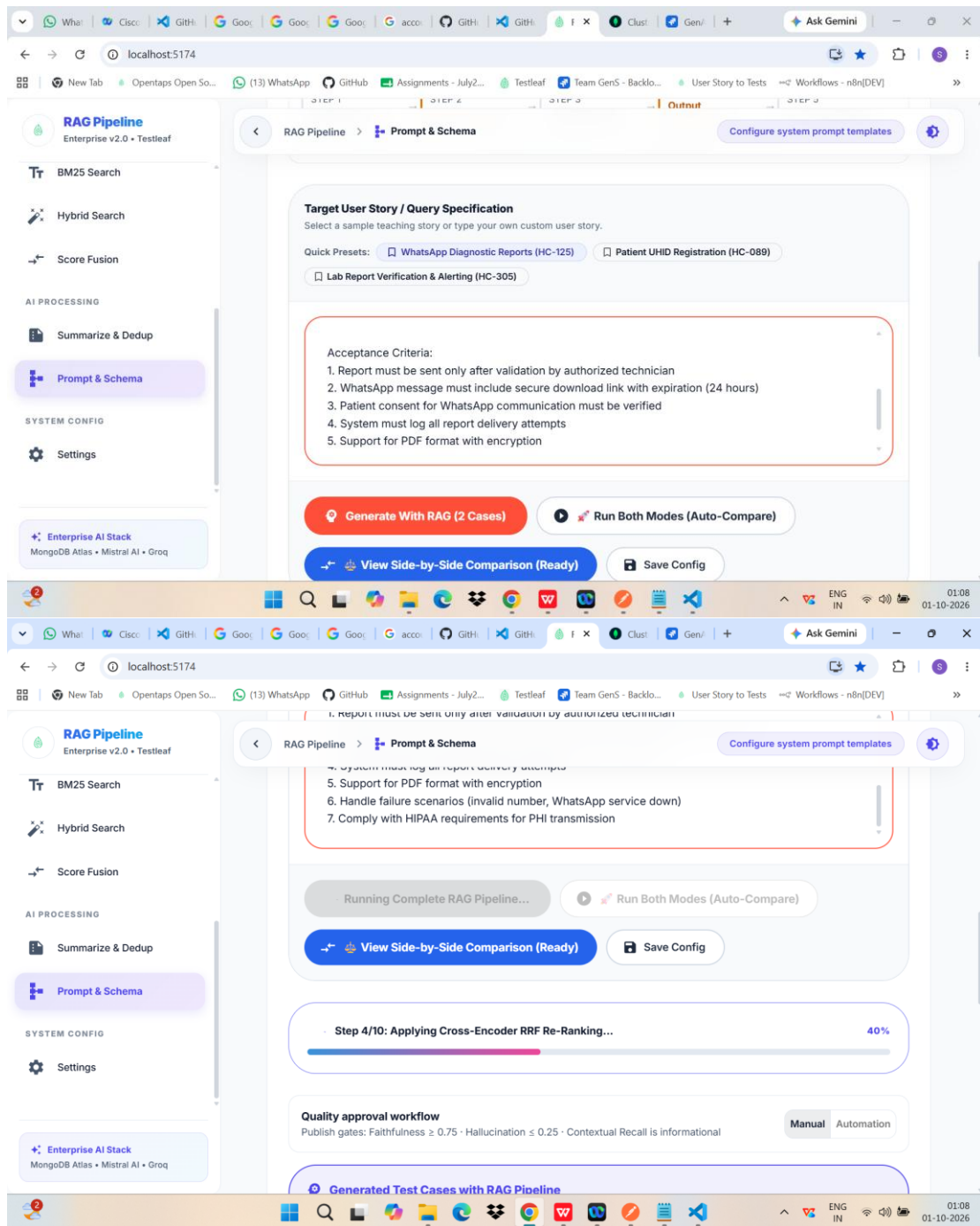
Alert System Status: All Good

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With RAG





The top screenshot displays the 'RAG Pipeline Enterprise v2.0 - Testleaf' interface. The left sidebar contains navigation options: BM25 Search, Hybrid Search, Score Fusion, AI PROCESSING (Summarize & Dedup), Prompt & Schema (selected), SYSTEM CONFIG (Settings), and Enterprise AI Stack (MongoDB Atlas, Mistral AI, Groq). The main content area shows the 'Prompt & Schema' configuration for 'RAG Pipeline'. It includes 'Acceptance Criteria' for WhatsApp reports, a 'Running Complete RAG Pipeline...' button, a 'Run Both Modes (Auto-Compare)' button, a 'View Side-by-Side Comparison (Ready)' button, and a 'Save Config' button. A 'DeepEval quality check 1/2: Positive - Successful WhatsApp Delivery of Encrypted PDF Report' is shown with a 90% progress bar. Below this, a 'Quality approval workflow' section lists publish gates: Faithfulness ≥ 0.75 , Hallucination ≤ 0.25 , and Contextual Recall is informational. The bottom section shows 'Generated Test Cases with RAG Pipeline'.

The bottom screenshot shows the 'Generated Test Cases with RAG Pipeline' page. It features a 'Evaluate Quality (DeepEval)' button and an 'Export CSV' button. The 'DeepEval Quality Review' section displays a table of test results:

Test Case	Faithfulness (≥ 0.75)	Contextual Recall (info)	Hallucination (≤ 0.25)	Gate
Positive - Successful WhatsApp Delivery of Encrypted PDF Report	N/A	0.000	0.000	Fail
Negative & Edge - Handling Missing Consent, Invalid Number, and Service Failure	1.000	0.000	0.000	Pass

Below the table, a warning message states: 'TC_6355: Faithfulness must be at least 0.75.' There are 'Approve & Publish' and 'Reject' buttons. At the bottom, it shows 'Generated Test Cases (2)', 'Retrieved MongoDB Context (8)', and 'Linked User Stories (10)'.

Enterprise v2.0 • Testleaf

BM25 Search

Hybrid Search

Score Fusion

AI PROCESSING

Summarize & Dedup

Prompt & Schema

SYSTEM CONFIG

Settings

Enterprise AI Stack

MongoDB Atlas • Mistral AI • Groq

Generated Test Cases with RAG Pipeline

RAG Pipeline

Prompt & Schema

Configure system prompt templates

Evaluate Quality (DeepEval)

Export CSV

DeepEval Quality Review

PUBLISHED

Test Case	Faithfulness (≥ 0.75)	Contextual Recall (info)	Hallucination (≤ 0.25)	Gate
Positive – Successful WhatsApp Delivery of Encrypted PDF Report	N/A	0.000	0.000	Fail
Negative & Edge – Handling Missing Consent, Invalid Number, and Service Failure	1.000	0.000	0.000	Pass

⚠️ TC_6355: Faithfulness must be at least 0.75.

✅ Published 2 test cases to MongoDB collection testcases and Jira.

Generated Test Cases (2)Retrieved MongoDB Context (8)Linked User Stories (10)

ID	Title	Module	Priority	Type	Precondition
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cloud.mongodb.com/v2/6a802f260e1ad45327638dbf#explorer/6a802f77669cc9eb12f3ed1f/RAG_testCases/testcases/find

ORGANIZATION Sam's Org - 2026-08-15PROJECT Project 0

Compare tiers

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

RAAG_testCases

FETCHED_JIRA_US

Generated_Approved_TestC...

Generated_Rejected_TestCa...

testcases

userStory

admin

local

testcases

Cluster0 > RAG_testCases > testcases

View monitoringVisualize Your Data

Documents6.4KAggregationsSchemaIndexes1Validation

Type a query: { field: 'value' } or Generate_query

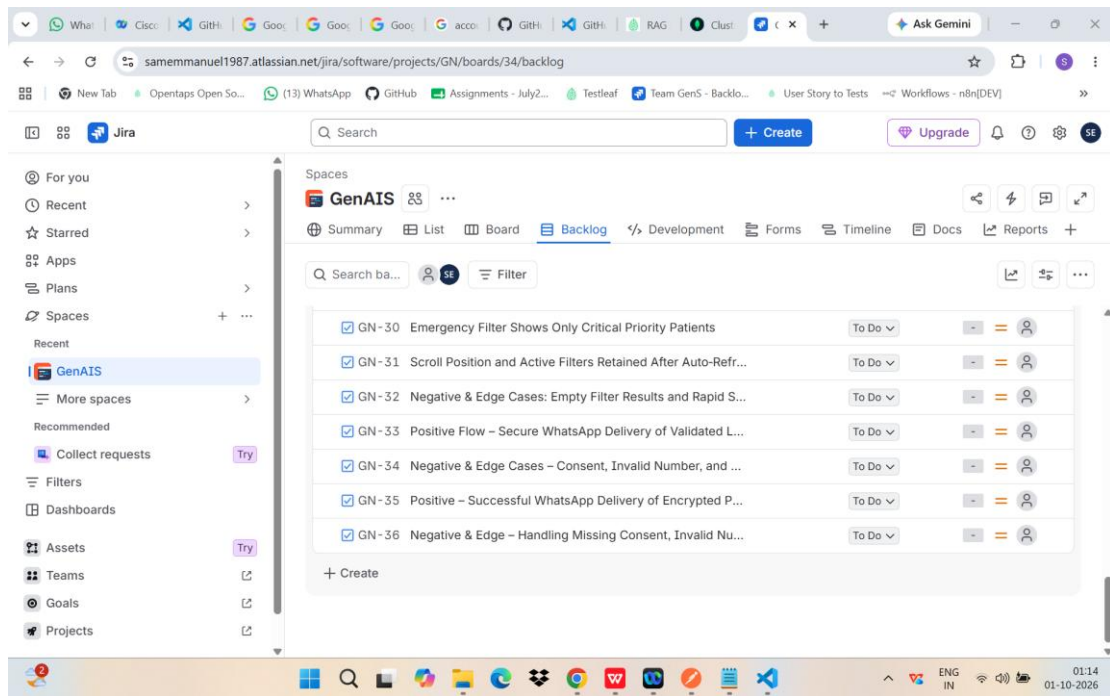
ExplainResetFindOptions

1001 - 100 of 6362

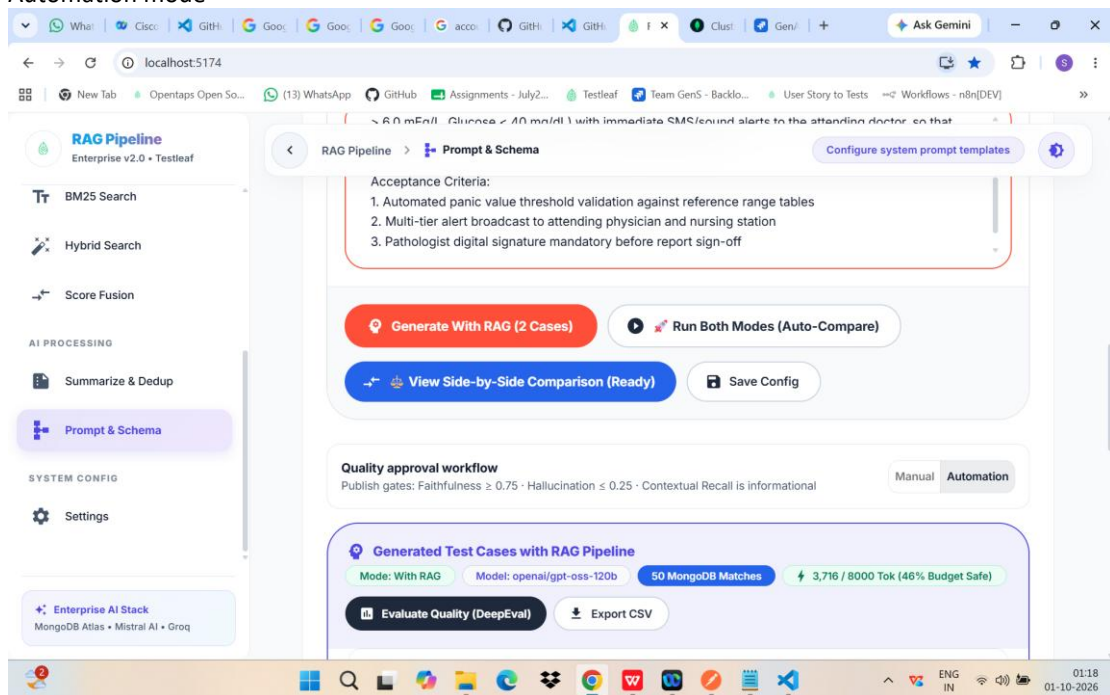
```
_id: ObjectId('6a820126e13415c9ef066fcb')
module: "Registration"
id: "TC_005"
preRequisites: "PRN/ERN Number"
title: "Conversion from PRN/ERN Reg to UHID"
description: "Verify Conversion from PRN/ERN Reg to UHID"
steps: "1.Launch the application
2.Login to the application with valid credentials.
3.Click on Patient Service Tab and select Registration from the drop down..."
expectedResults: "Should get convert successfully from PRN/ERN Reg to UHID"
automationManual: "Automated"
priority: "P1"
```

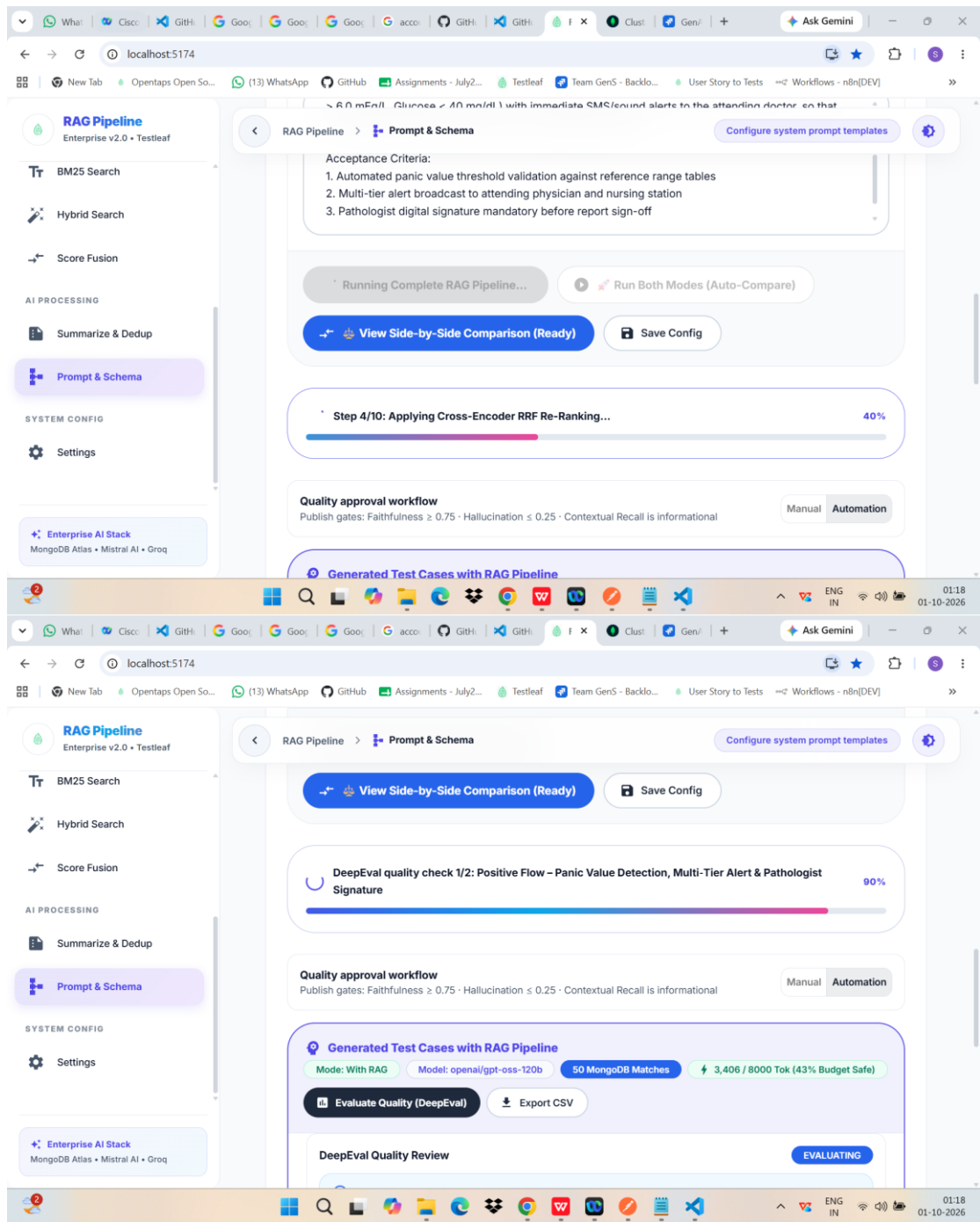
Alert System Status: All Good

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Automation mode





The screenshot displays the RAG Pipeline application interface. The top navigation bar includes links to various services like Ciscr, GitHub, Google, and a search bar. The main content area is titled "RAG Pipeline" and "Enterprise v2.0 • Testleaf". It features a sidebar with navigation options: BM25 Search, Hybrid Search, Score Fusion, AI PROCESSING (Summarize & Dedup), Prompt & Schema (highlighted), SYSTEM CONFIG (Settings), and Enterprise AI Stack (MongoDB Atlas, Mistral AI, Groq). The main content area shows "Generated Test Cases with RAG Pipeline" with a status bar indicating "Mode: With RAG", "Model: openai/gpt-oss-120b", "50 MongoDB Matches", and "3,406 / 8000 Tok (43% Budget Safe)". Below this is a "Evaluate Quality (DeepEval)" button and an "Export CSV" button. The "DeepEval Quality Review" section contains a table with columns: Test Case, Faithfulness (≥ 0.75), Contextual Recall (info), Hallucination (≤ 0.25), and Gate. The table lists two test cases: "Positive Flow – Panic Value Detection, Multi-Tier Alert & Pathologist Signature" (Pass) and "Negative & Boundary Cases – No Alert for Non-Panic Values, Failed SMS Handling, Duplicate Alert Suppression" (Fail). A red alert box at the bottom states: "TC_6356: Faithfulness must be at least 0.75. TC_6356: Hallucination must be at most 0.25." The bottom status bar shows "Generated Test Cases (2)", "Retrieved MongoDB Context (5)", and "Linked User Stories (10)".

Generated Test Cases with RAG Pipeline

Mode: With RAG Model: openai/gpt-oss-120b 50 MongoDB Matches 3,406 / 8000 Tok (43% Budget Safe)

Evaluate Quality (DeepEval) Export CSV

DeepEval Quality Review

Test Case	Faithfulness (≥ 0.75)	Contextual Recall (info)	Hallucination (≤ 0.25)	Gate
Positive Flow – Panic Value Detection, Multi-Tier Alert & Pathologist Signature	1.000	0.000	0.000	Pass
Negative & Boundary Cases – No Alert for Non-Panic Values, Failed SMS Handling, Duplicate Alert Suppression	N/A	N/A	N/A	Fail

⚠️ TC_6356: Faithfulness must be at least 0.75. TC_6356: Hallucination must be at most 0.25.

Generated Test Cases (2) Retrieved MongoDB Context (5) Linked User Stories (10)